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A large-scale circular RNA profiling reveals universal molecular mechanisms responsive to drought stress in maize and Arabidopsis

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SUMMARY

Drought is a major abiotic stress that threatens global food security. Circular RNAs (circRNAs) are endogenous RNAs. How these molecules influence plant stress responses remains elusive. Here, a large-scale circRNA profiling identified 2174 and 1354 high-confidence circRNAs in maize and Arabidopsis, respectively, and most were differentially expressed in response to drought. A substantial number of drought-associated circRNA-hosting genes were involved in conserved or species-specific pathways in drought responses. Comparative analysis revealed that circRNA biogenesis was more complex in maize than in Arabidopsis. In most cases, maize circRNAs were negatively correlated with sRNA accumulation. In 368 maize inbred lines, the circRNA-hosting genes were enriched for single nucleotide polymorphisms (SNPs) associated with circRNA expression and drought tolerance, implying either important roles of circRNAs in maize drought responses or their potential use as biomarkers for breeding drought-tolerant maize. Additionally, the expression levels of circRNAs derived from drought-responsible genes encoding calcium-dependent protein kinase and cytokinin oxidase/dehydrogenase were significantly associated with drought tolerance of maize seedlings. Specifically, Arabidopsis plants overexpressing circGORK (Guard cell outward-rectifying K^+-channel) were hypersensitive to abscisic acid, but insensitive to drought, suggesting a positive role of circGORK in drought tolerance. We report the transcriptomic profiling and transgenic studies of circRNAs in plant drought responses, and provide evidence highlighting the universal molecular mechanisms involved in plant drought tolerance.

Keywords: circular RNAs, transcriptome, drought stress, association analysis, Arabidopsis (*Arabidopsis thaliana*), maize (*Zea mays*).

INTRODUCTION

Drought is one of the main environmental factors damaging agriculture worldwide. Studying how plants, especially crops, respond to drought stress is a fundamental scientific question and discoveries will be beneficial for breeding drought-resistant crops. Over the past decades, great advances have been made in deciphering the genetic and molecular bases of plant drought responses. The phytohormone abscisic acid (ABA) is essential for plant stress responses. Under drought stress, plants increase the production of ABA, which helps to promote stomatal closure and prevent water loss (Pantin *et al.*, 2013; Munemasa *et al.*, 2015). Numerous studies have uncovered regulation of drought responses by the ABA signaling pathway. This involves a group of key players, including PYL/RCAR ABA receptors, clade A PP2C phosphatases, SnRK2 kinases and a number of transcriptional factors (TF) such as bZIP, NAC, WRKY etc. (Fujita *et al.*, 2011). Drought also promotes the production of reactive oxygen species (ROS) that can damage cells (Miller *et al.*, 2010). To survive drought, plants have developed enzymatic and non-enzymatic antioxidative defense mechanisms to scavenge ROS (Gill and Tuteja, 2010). Additionally, non-coding RNAs, such as microRNA (miRNA) and long non-coding RNA (lncRNA) have also been reported to play important roles in the regulation of plant drought responses (Di *et al.*, 2014; Ferdous *et al.*, 2015).

High-throughput sequencing technologies have shown that eukaryotes contain large numbers of circular RNAs (circRNAs), which are endogenously processed from their cognate linear RNAs (Barrett and Salzman, 2016). Recent advances reveal that circRNAs have essential functions in animals. For example, *ciRS-7*, a circRNA conserved in mammals, has more than 70 potential miR-7 binding sites, and serves as a sponge regulating miRNA function and brain development (Hansen *et al.*, 2013; Memczak *et al.*, 2013). A class of regulatory circRNAs, named exon–intron circRNAs (EIciRNAs), has been shown to regulate gene transcription (Chen *et al.*, 2015; Li *et al.*, 2015). In the fly brain, the abundant circRNAs of the *mbl* gene bind and sequester MBL protein, in turn autoregulating MBL protein levels (Ashwal-Fluss *et al.*, 2014). Additionally, several research groups have reported that some circRNAs can encode functional proteins (Legnini *et al.*, 2017; Pamudurti *et al.*, 2017). Together, these studies suggest broad and important roles for circRNAs in the regulation of animal development.

Thought to be produced by mis-splicing, circRNA molecules were first identified in human cells more than 20 years ago (Hsu and Coca-Prados, 1979; Cocquerelle *et al.*, 1993). However, recent advances have revealed that the biogenesis of circRNAs is complex and regulated by both *cis*- and *trans*-acting factors. Studies from humans and flies suggest that inverted repeats, which are usually derived from transposons, are required for circRNA production (Jeck *et al.*, 2013; Memczak *et al.*, 2013; Liang and Wilusz, 2014). We recently reported that the LINE1-like transposable elements (LLEs) and their related reverse complementary pairs (LLERCPs) were significantly enriched in the regions flanking maize circRNAs, suggesting the involvement of these repeated sequences in plant circRNA production (Chen *et al.*, 2018). The RNA binding protein muscleblind, heterogeneous nuclear ribonucleoproteins (hnRNPs) as well as SR proteins are necessary for circRNA production in *Drosophila* (Ashwal-Fluss *et al.*, 2014; Kramer *et al.*, 2015). Similarly, in mammals, RNA-binding proteins, such as quaking (QKI) and ADAR family proteins, are essential for RNA circularization (Conn *et al.*, 2015; Barrett and Salzman, 2016). Recent studies have shown that the immune factors NF90/NF110 regulate circRNA biogenesis and function in response to virus infection (Li *et al.*, 2017). Even though thousands of circRNAs have been identified with deep RNA-seq and bioinformatic tools in several plant species, such as Arabidopsis, rice, tomato and soybean (Lu *et al.*, 2015; Ye *et al.*, 2015; Conn *et al.*, 2017; Tan *et al.*, 2017; Zhao *et al.*, 2017), their roles in regulation of plant stress responses remain largely unknown.

In this study, we subjected maize and Arabidopsis plants to various levels of drought. Circular and linear transcriptomes of these plants were obtained by deep RNA-seq. We detected thousands of circRNAs, among which hundreds were found to be differentially expressed in both plant species after drought stress, indicating that circRNAs are effective indicators of plant drought responses. Comparative analyses of circular and linear transcripts revealed the shared and species-specific pathways in drought responses of maize and Arabidopsis. Association analyses in a large maize population and transgenic studies in Arabidopsis uncovered links between circRNA expression and drought tolerance, suggesting that circRNAs either play critical roles in plant drought response or can be used as effective biomarkers in genetic improvement of crop drought tolerance.

RESULTS

Experimental design and plant physiological responses to drought

To better understand how drought regulates the growth of maize and Arabidopsis plants, we grew the plants in glasshouses and subjected them to various levels of drought stress (Experimental procedures). Briefly, wild-type maize B73 seeds or Arabidopsis Columbia (Col) seeds were sown four per pot and allowed to grow under well watered (WW) conditions [*c.* 65–75% soil moisture for maize (MWW) and *c.* 80–100% soil moisture for Arabidopsis (AWW)] for approximately 15 days for maize and 21 days for Arabidopsis, respectively. At this point, some plants continued to be watered, watering of other plants (drought-treated, DT) ceased until their soil moisture reached a designated percentage, then this soil moisture level was maintained by daily watering until harvesting. The soil moistures for MDT1 maize plants were 40–45%, 30–35% for MDT2, 20–25% for MDT3, 10–15% for MDT4, while the soil moistures for Arabidopsis ADT1 plants were 50–55%, 40–45% for ADT2, 30–35% for ADT3, 20–25% for ADT4, 10–15% for ADT5 (Figure 1a,b). We observed that after ceasing watering it took 3 days to reach MDT1 and 6 days to reach MDT4, whereas reaching ADT1 and ADT5, took 4 and 8 days, respectively (Figure 1a,b). Maize plants were harvested on treatment day 7, when the seedlings had *c.* 4 true leaves, while Arabidopsis plants were harvested on treatment day 9 and had *c.* 10 rosette leaves.

Growth analyses showed that MDT1 and MDT2 maize plants were similar to those of MWW plants, with well expanded leaves, whereas the MDT3 and MDT4 plants showed obvious wilted leaves caused by drought (Figure 1c). For Arabidopsis, ADT1, ADT2 and ADT3 plants had expanded leaves, similar to AWW plants, although they were smaller overall than AWW plants (Figure 1d). ADT4 and ADT5 plants had wilted leaves indicating stronger effects of drought stress on their growth (Figure 1d). Biomass analyses showed that MDT1, MDT2, MDT3 and

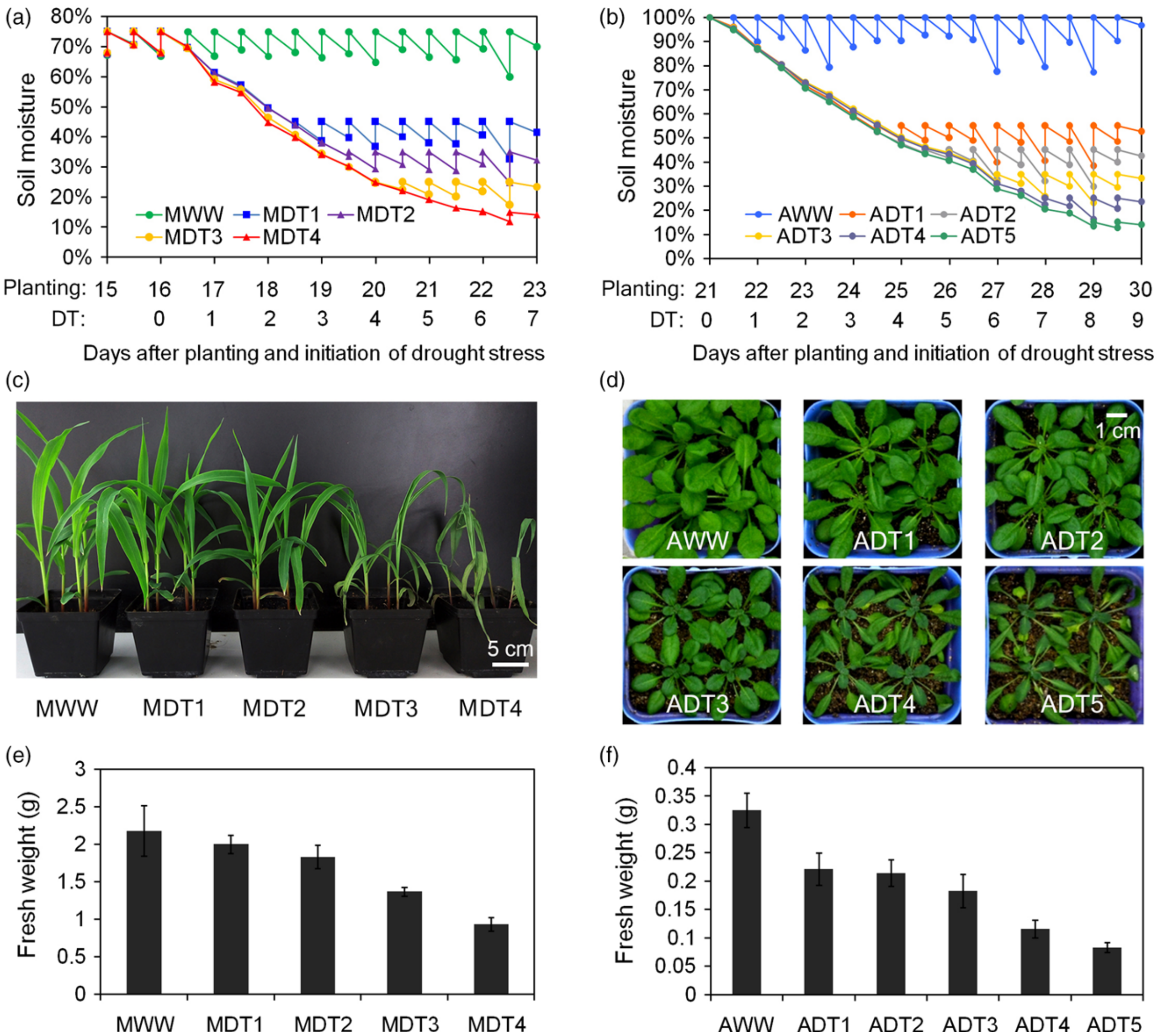

Figure 1. Physiological responses of maize and Arabidopsis plants to drought stress.
(a, b) Soil moisture dynamics during drought stress in maize (a) and Arabidopsis plants (b). MWW, maize well watered; MDT, maize drought-stressed; AWW, Arabidopsis well watered; ADT, Arabidopsis drought-stressed. Plants were sampled on the last day of drought stress.
(c, d) Morphological phenotypes of maize (c) and Arabidopsis (d) plants subjected to various levels of drought stress. Photographs were taken on the same day as harvesting. Bar = 5 cm in (c) and 1 cm in (d)
(e, f) Fresh weights of maize (e) and Arabidopsis (f) subjected to various levels of drought stress. Bars indicate ± standard deviation (SD) ($n \geq 3$).

MDT4 maize plants lost increasing amounts of fresh weight following drought stress (Figure 1e). Based on their fresh weights, MDT1 and MDT2 plants were only slightly affected by drought stress, while MDT3 and MDT4 were severely affected (Figure 1e). Similarly, Arabidopsis ADT1, ADT2, ADT3, ADT4 and ADT5 plants also lost increasing amounts of their fresh weight after drought stress, where ADT1, ADT2 and ADT3 were slightly affected, while ADT4 and ADT5 plants were strongly affected (Figure 1f). Taken together, all these observations indicated that drought negatively affected the growth of both maize and Arabidopsis plants with a direct correlation between the severity of drought and plant growth. Intriguingly, Arabidopsis plants were more sensitive to drought than maize plants grown in the same conditions. All Arabidopsis plants grown at similar soil moisture levels lost more fresh weight compared with their respective WW plants than maize. For example, ADT2 plants lost more fresh weight than MDT1 plants, which were both grown at 40–45% soil moisture. Similarly, Arabidopsis lost more fresh weight than maize in all of the pairs grown at similar soil moisture levels, such as ADT3-MDT2 (30–35% soil moisture), ADT4-MDT3 (20–25% soil moisture) and ADT5-MDT4 (10–15% soil moisture) pairs (Figure 1e,f).

Circular and linear transcripts are effective indicators of plant response to drought stress

We performed a comprehensive transcriptomic profiling to dissect molecular mechanisms associated with plant drought responses. Our profiling included mRNA sequencing (mRNA-seq) and circRNA sequencing (circRNA-seq) on maize and Arabidopsis leaves subjected to various drought treatments (Experimental procedures; Figure S1). We chose MWW, MDT2, MDT3 and MDT4 maize plants for transcriptomic analysis. MDT1 plants were removed from further analysis because their morphology and fresh weight resembled MWW plants (Figure 1). Similarly, as ADT1 and ADT2 Arabidopsis plants were very similar (Figure 1), we only used AWW, ADT2, ADT3, ADT4 and ADT5 plants for transcriptomic sequencing. Maize and Arabidopsis leaves were harvested at treatment days 7 and 9, respectively, for RNA isolation and transcriptomic profiling by RNA-seq.

Over 1.1 and 1.4 billion circRNA-seq reads were obtained from maize and Arabidopsis, respectively (Tables S1 and S2). Additionally, approximately 400 million mRNA-seq reads were collected in both species with an average of 40 million clean reads in each replicate (Tables S3 and S4). All of these circRNA-seq and mRNA-seq data were subjected to a series of stringent quality controls and mapped to reference genome B73 for maize or reference genome Col for Arabidopsis. Hundreds of circRNAs as well as tens of thousands of linear transcripts were detected that exhibited characteristic expression patterns across different drought treatments in maize and Arabidopsis (Tables S5–S8). Principle component analyses (PCA) of both circRNA and linear transcripts in maize and Arabidopsis showed that the replicates of every treatment clustered together, indicating the high reliability of the biological samples (Figure S2). The replicates for each drought treatment clustered separately from the other treatments according to components 1 and 2, which explained the majority of the transcriptomic variation in both circRNA and linear transcripts (Figure S2). This profiling showed that each drought treatment resulted in dramatic changes in the transcriptomes of both maize and Arabidopsis.

For circRNAs, only high-confidence circRNAs with at least two back-spliced junction reads were utilized for the further dissection of molecular responses to drought treatments. In total, 2174 circRNAs were detected in maize and 1354 circRNAs in Arabidopsis, and more exonic than intronic or intergenic circRNAs were detected in both plant species (Figure 2). The number of detectable circRNAs varied

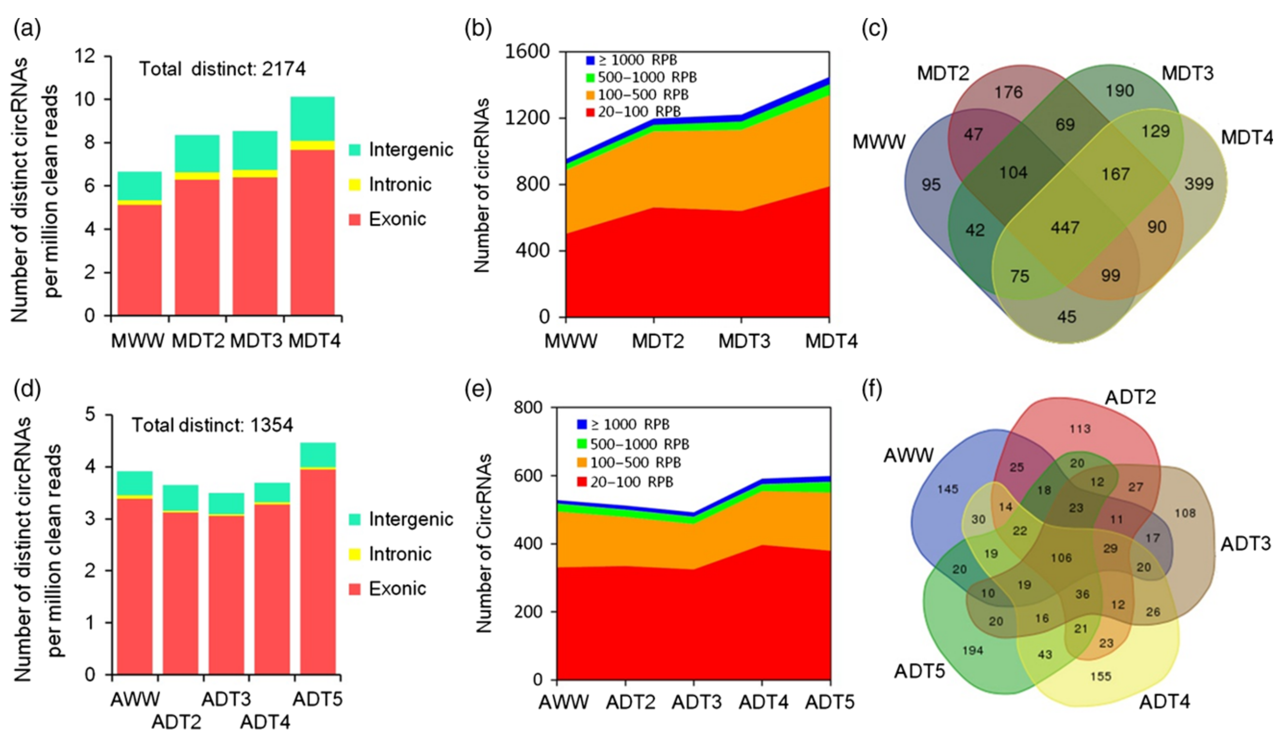

Figure 2. Accumulation of circRNAs in drought-stressed maize and Arabidopsis plants.
(a, d) The amounts of distinct circRNAs detected in maize (a) and Arabidopsis (d) subjected to various levels of drought stress. circRNAs detected in both replicates were selected as 'high-confidence' (2174 and 1354, respectively, in maize and Arabidopsis) for further analysis.
(b, e) The amounts of circRNAs with different expression levels in maize (b) and Arabidopsis (e) plants subjected to various levels of drought stress.
(c, f) Venn diagrams showing the overlaps of circRNAs expressed in maize (c) and Arabidopsis (f) subjected to various levels of drought stresses.

across different drought treatments in both species, suggesting a treatment-specific pattern of circRNA expression in plants (Figure 2a,d). Interestingly, the relative number of circRNAs, normalized by the total number of circRNA-Seq reads, increased with drought in maize, reaching its highest level in MDT4 (Figure 2a). By classifying these detectable circRNAs according to their expression levels, it is worth noting that the increased number of detectable circRNAs is mainly due to the increase in circRNAs expressed at low levels (<100 reads per billion (RPB)) compared with the stable number of moderately (100–500 RPB) and highly expressed ones (≥500 RPB) (Figure 2b). This suggests that plants use circRNAs as an on-off switch or in a specific expression pattern responsive to drought stress. Accordingly, only 447 out of 2174 circRNAs were simultaneously detected in all drought treatments in maize (Figure 2c). In Arabidopsis, the relative number of circRNAs decreased 10.7% from AWW to ADT3, but then increased dramatically in ADT5, which had the most of any treatment (Figure 2d). The number of circRNAs expressed at low and moderate levels increased, while the levels of highly expressed circRNAs were consistent with the drought treatments (Figure 2e). Similarly, a small proportion (106/1354) of circRNAs was detected in all treatments in Arabidopsis (Figure 2f). As a counterpart of circRNAs, linear transcripts exhibited coordinated variations in expression levels across different drought treatments in both maize and Arabidopsis (Figure S3). All these results show that circular and linear transcripts are effective indicators of genome-wide molecular landscapes of plant responses to drought stress.

We compared the expression levels of circRNA-hosting genes and the other coding genes not producing circRNAs in maize and Arabidopsis treated with various levels of drought. The results showed that the expression levels of circRNA-hosting genes were significantly higher than those of the other coding genes (Figure S4). These results suggested that the highly expressed genes in general tended to generate circRNAs in both species.

Validation of circular RNAs and linear transcripts

To validate the circular RNAs detected by RNA sequencing, we chose a number of circRNAs from maize and Arabidopsis, respectively, for polymerase chain reaction (PCR) amplification. A series of convergent and divergent primers was used in the analysis (Table S9). A pair of convergent primers was designed to amplify the linear DNA fragments when cDNA or genomic DNA was used as templates in the PCR reactions. By contrast, the divergent primers, in which the reverse primers were located upstream of the forward primers, were designed to amplify cDNA fragments from circRNA-derived cDNAs. As a control, the PCR reaction with divergent primers would not generate DNA fragments when genomic DNA was used as template. DNA sequencing of the PCR products was performed to further confirm the back-spliced junctions of the circRNAs (Figure 3a–f). In total, 73 circRNAs, 42 from maize and 31 from Arabidopsis, respectively, were validated by PCR reactions (Table S9). Sequencing analyses were used to confirm the junction sites of all PCR products. Sixty-eight of the 73 circRNAs (*c.* 93%) were validated in our experiments (Figures 3 and S5; Table S9). We next validated the expression levels of circRNAs and linear transcripts with quantitative RT-PCR (QPCR), and the results were consistent with the sequencing data (Figure S6). These results indicate the accuracy of our analyses detecting maize and Arabidopsis circRNA and linear transcripts.

Correlations between circular and linear transcripts in drought response

Differential expression analyses identified 1843 circRNAs and 8681 linear transcripts that were differentially expressed in pairs of drought treatments in maize, and comparable numbers of differentially expressed circRNAs (1283) and linear transcripts (9322) in Arabidopsis (Tables S10–S13). Hierarchical clustering of variation in expression levels showed that accumulation of circRNA and linear transcripts either increased or decreased with different drought treatments in both maize and Arabidopsis, indicative of synergistic molecular mechanisms for circRNAs and linear transcripts responsive to drought stress in plants (Figure 4a–d). We next compared the magnitude of the changes in circRNA and mRNA levels after drought stresses. Surprisingly, most of the circRNAs showed much greater changes than those of mRNAs in response to drought (Figure S7), suggesting that circRNAs were more sensitive to drought than mRNAs.

Genomic locus coincidence analysis in maize showed that a substantial proportion (357/1100) of loci which differentially expressed circRNAs (MDECs) were located in gene regions which also differentially expressed linear transcripts (MDELs) (Figure 4e). The proportion of the same type of loci in Arabidopsis was 446/883 (Figure 4f). Clustering analyses of circRNAs and linear transcripts divided these differentially expressed transcripts into 20 groups in maize and Arabidopsis, respectively (Figure 4g,i). In particular, genes with both differentially expressed circRNAs and linear transcripts (DEGs) that either increased or decreased under different drought treatments were significantly enriched in both maize and Arabidopsis (Figure 4h,j). However, those with opposite variation in expression levels of circRNAs and linear transcripts were significantly depleted in maize and Arabidopsis in response to drought (Figure 4h,j). All these results suggested a similar variation pattern in gene expression responding to drought stress in both maize and Arabidopsis, implicating possible conserved molecular mechanisms for drought responses in plants.

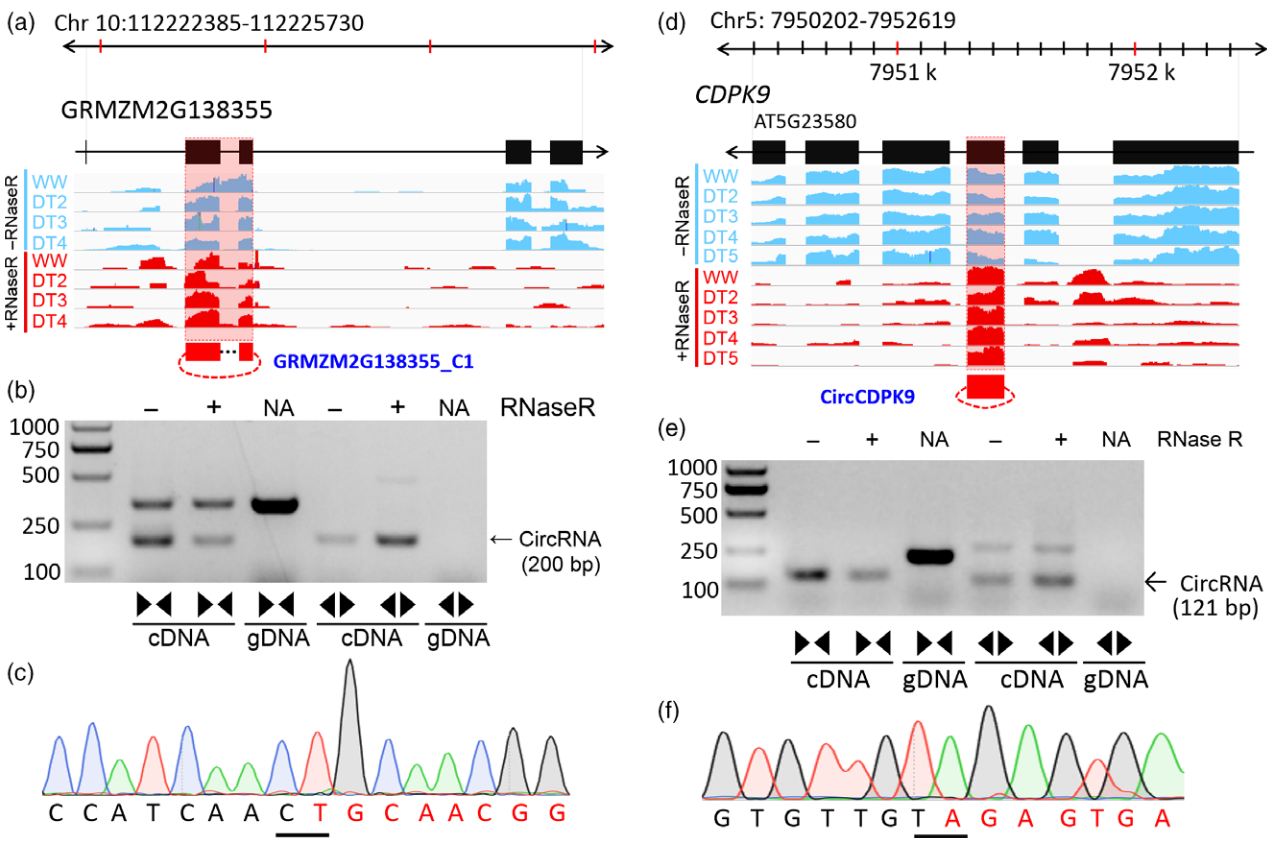

Figure 3. Validation examples of circRNA production.
(a, d) The derivation of maize circRNA GRMZM2G138355_C1 (a) and Arabidopsis circCDPK (d), the gene structures and physical positions of these genes, the RNA-seq reads mapped to these two loci with or without RNase *R* treatments were shown below. circRNAs are shown as red circles.
(b, e) The PCR reactions with divergent and convergent primers using different templates showing the production of GRMZM2G138355_C1 (b) and circCDPK (e).
(c, f) Sequencing confirmation of the junction site of GRMZM2G138355_C1 (c) and circCDPK (f).

Evolution of plant circRNAs in drought response

To gain insight into the evolution of plant circRNAs, we compared the circRNAs of Arabidopsis and maize which were differentially expressed after drought stress. Of all the 1100 circRNA-hosting genes in maize, 482 (43.82%) were orthologs to 401 (47.97% of 836) circRNA-hosting genes in Arabidopsis (Figure 5a). Therefore, maize and Arabidopsis shared a large proportion of hosting genes to generate circRNAs in drought responses. Interestingly, we found that the orthologs circRNAs tend to be enriched in drought-induced patterns in both maize and Arabidopsis (Figure S8). Kyoto Encyclopedia of Genes and Genomes (KEGG) analyses showed that the shared hosting genes in maize and Arabidopsis were enriched in 22 pathways, with ‘peroxisome’, ‘carbon fixation in photosynthetic organisms’, ‘carbon metabolism’, ‘nitrogen metabolism’ and ‘metabolic pathways’ as the top five (Figure 5b).

We took a closer look at some drought-responsive pathways enriched with circRNA-hosting genes. In the ABA signaling pathway, one gene from maize and ABI2 from Arabidopsis, which both encode clade A PP2C proteins, were detected; OST1 (SnRK2 kinase) and ABF3 (bZIP transcription factor) from Arabidopsis were also mapped to this pathway (Figure S9a). In the peroxisomal biogenesis pathway, a number of matrix proteins and import proteins from both maize and Arabidopsis were detected (Figure S9b). For example, four genes from maize and six genes from Arabidopsis involved in fatty acid oxidation, antioxidant system and amino acid metabolism were detected. Two other maize genes detected in this pathway were responsible for either matrix protein import or retinol metabolism. Another detected gene from Arabidopsis was responsible for ROS metabolism (Figure S9b). Four genes from maize and three genes from Arabidopsis responsible for the C4-dicarboxylic acid cycle were detected in the carbon fixation pathway, together with four genes from maize and six genes from Arabidopsis responsible for carbon fixation (Figure S9c). These observations implied that circRNA production is generally conserved between maize and Arabidopsis. We also observed species-specific KEGG pathways for the circRNA-hosting genes in both maize and Arabidopsis (Figure 5c,d). Intriguingly, among the 20 enriched KEGG pathways of Arabidopsis-specific hosting

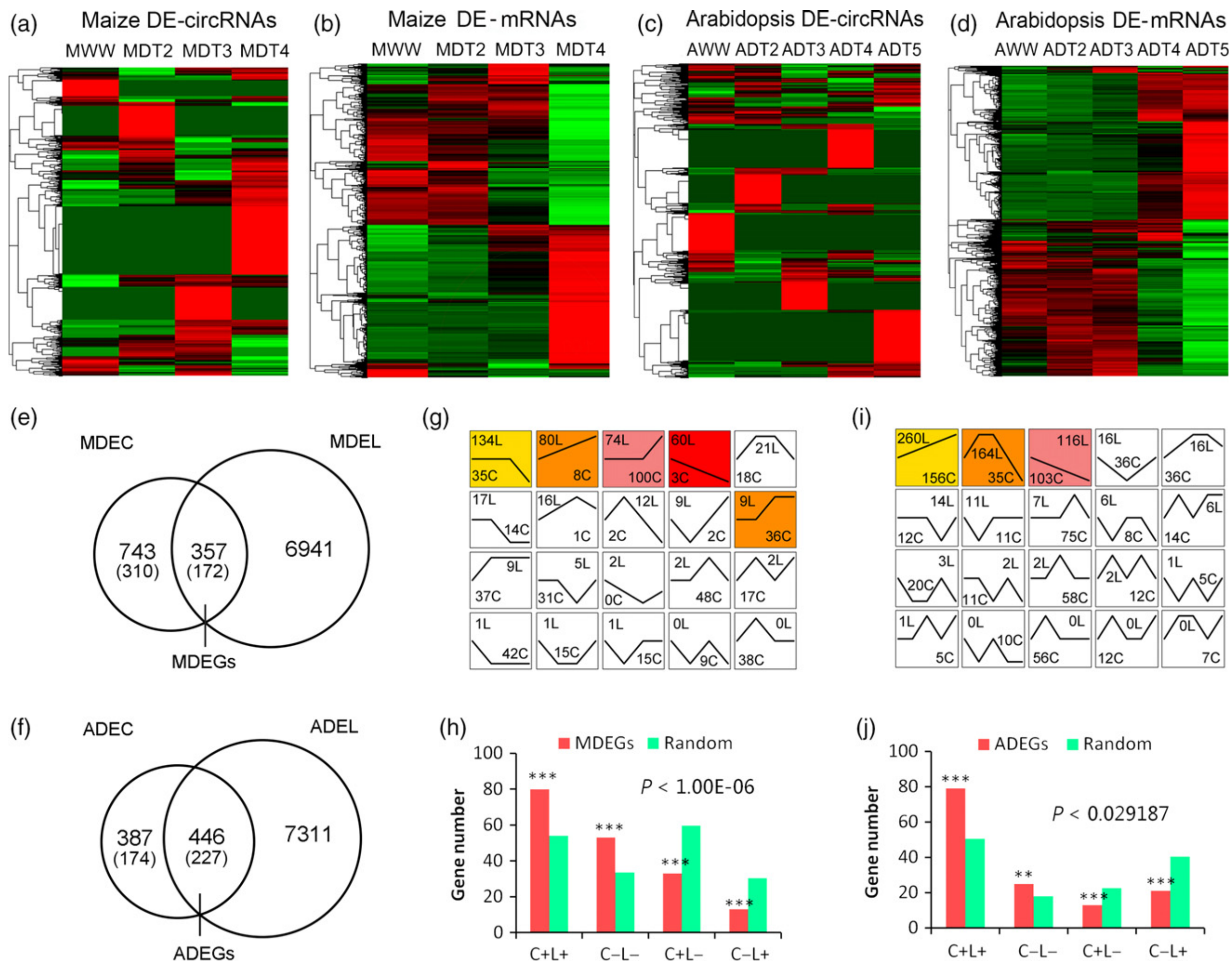

Figure 4. Variation in expression of plant circular and linear transcripts in response to drought stress.
(a, c) Clustered heatmaps of differentially expressed (DE) circRNAs in drought-stressed maize (a) and Arabidopsis (c) plants. The RNAs were classified according to the Pearson correlation. The numerical data represent $\log_{10}$-transformed mean RPB of two replicates.
(b, d) Clustered heatmaps of DE linear mRNAs in drought-stressed maize (b) and Arabidopsis (d) plants. The RNAs were classified according to the Pearson correlation. The numerical data represent $\log_{10}$-transformed mean TPM of two replicates.
(e, f) Venn diagrams showing the overlapped maize genes (e, MDEG) and Arabidopsis genes (f, ADEG) producing differentially expressed circular (DEC) and linear (DEL) transcripts in response to drought, respectively. The amounts of orthologs circRNAs were shown in the brackets.
(g, i) Expression patterns of linear RNAs (L) and circular RNAs (C) of DEGs in maize (g) and Arabidopsis (i). Trend lines depict the changes of transcripts in plants subjected to various levels of drought. The amounts of RNA species related a certain group are indicated. Colors indicate the enriched patterns.
(h, j) Comparison of gene numbers showing expression patterns of CxLx (C+L+, C-L−, C+L− or C-L+) in DEGs and those from random selections in maize (h) and Arabidopsis (j). '+' and '−', upregulation or downregulation of a transcript in response to drought, respectively. Simulation analyses of randomly selected genes expressing either DECs or DELs were used as controls. p denotes the maximum probability among the four simulation analyses (n = 1 000 000).

genes, eight were also detected in the pathways of the shared genes, and among all the eight enriched KEGG pathways of maize-specific hosting genes, one was detected in the pathways of the shared genes (Figure 5c,d). Therefore, besides the conserved mechanisms, plants also adopt some species-specific mechanisms in circRNA production in response to drought.

An interesting finding was that the maize drought-responsive circRNAs outnumbered those of Arabidopsis (1100 versus 836) (Figure 5a). It has been reported that the flanking introns are important for circRNA generation (Dang *et al.*, 2016). We compared the flanking introns of circRNAs in maize and Arabidopsis. The results showed that the lengths of the flanking introns of circRNAs in each plant were in general longer than the average introns in their respective genomes (t-test, $P < 7.9E-46$ in maize and $P < 5.3E-34$ in Arabidopsis) (Figure 5e). In a given plant species, there was no significant difference in the flanking-intron lengths of the circRNAs generated from the shared and species-specific hosting genes (Figure 5e).

It has been reported that the repeated sequences flanking the circRNAs are important for circRNA formation in

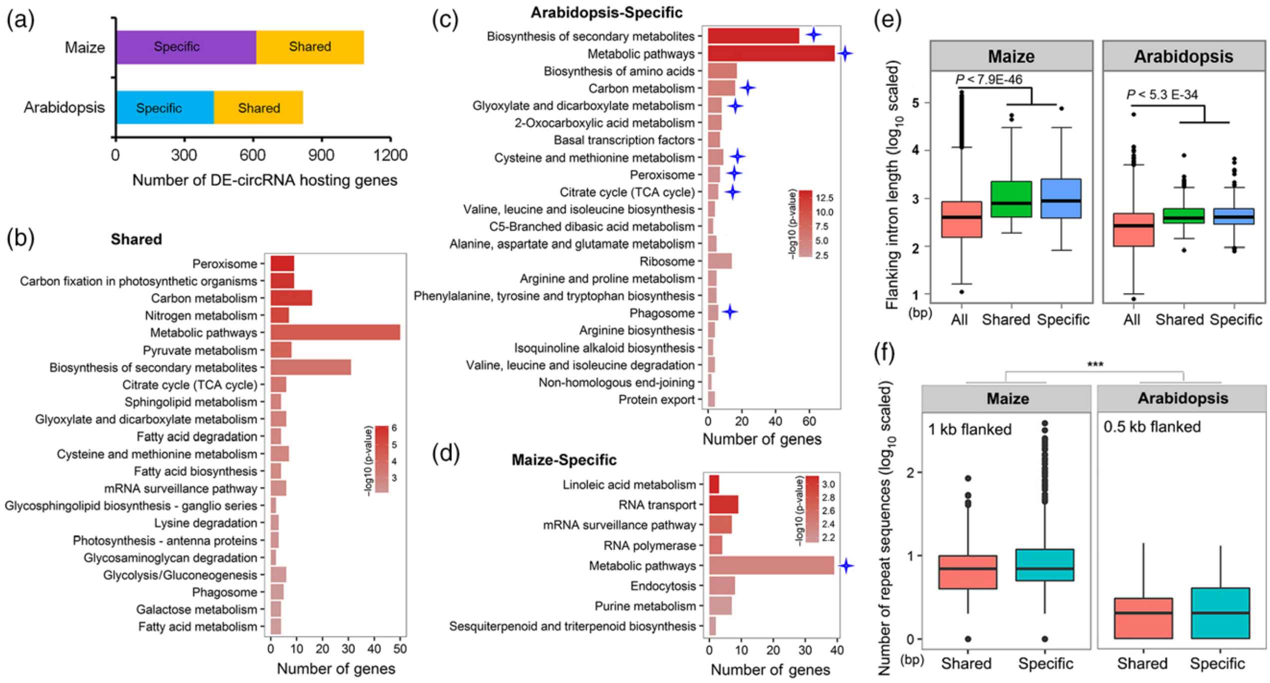

Figure 5. Comparative analyses of circRNAs and their hosting genes in Arabidopsis and maize.
(a) Numbers of DE-circRNA-hosting genes in maize and Arabidopsis. 'Shared', orthologous genes in maize and Arabidopsis; 'Specific', genes only exist in maize or Arabidopsis.
(b–d) KEGG pathways enriched with the shared (b) and Arabidopsis-specific (c) or maize-specific (d) circRNA-hosting genes in response to drought stress. Stars indicate the pathways of both species-specific and shared genes.
(e) Comparisons of the lengths of introns flanking circRNA junction sites between 'Shared' and 'Species-specific' hosting genes, with lengths of all introns in maize or Arabidopsis as backgrounds.
(f) Comparison of the numbers of repeated sequences in regions flanking circRNA junction sites between 'Shared' and 'Species-specific' hosting genes. Student's *t*-tests were performed between different groups: ***$P < 0.001$.

animals (Jeck *et al.*, 2013; Memczak *et al.*, 2013; Liang and Wilusz, 2014). We have recently reported that TE-related repeated sequences in the flanking regions of circRNAs play important roles in maize circRNA formation (Chen *et al.*, 2018). We therefore counted and compared the numbers of repeated sequences in the flanking introns of circRNAs in maize and Arabidopsis. One kb and 0.5 kb sequences flanking the circRNAs in maize and Arabidopsis, respectively, were counted, because the median lengths of the flanking introns in maize and Arabidopsis were about 1 kb and 0.5 kb, respectively. We observed many more repeated sequences in maize introns than in Arabidopsis introns (*t*-test, $P < 0.001$) (Figure 5f).

Effects of circRNAs on sRNA accumulation

Small RNAs (sRNAs), which are produced from transposable elements (TEs), microRNA loci, inverted repeats and other loci, play pivotal roles in regulation of plant development and stress responses through machineries of transcriptional gene silencing (TGS) or post-transcriptional gene silencing (PTGS) (Huang *et al.*, 2011). To know how circRNAs affect sRNA production, we investigated the interactions between circRNA expression and sRNA accumulation during drought responses. A previous report has identified genome-wide responses of sRNAs to drought in B73 seedlings (Li *et al.*, 2013). We used their sRNA-seq data in our analyses. We reanalyzed these data, and compared the expression levels between the TE-derived sRNAs mapped to circRNAs (circRNA-mapped) and those could not (circRNA-nonmapped) (Figure 6a). When they were 21- or 22-nt-long, the expression of circRNA-nonmapped sRNAs were significantly higher than those mapped to circRNAs, no matter with or without drought stress (on average, 1.2-fold increase for 21nt-sRNAs and 1.5-fold increase for 22nt-sRNAs) (Figure 6b). Interestingly, the expression of circRNA-nonmapped 24nt-sRNAs was slightly but significantly decreased as compared with those circRNA-mapped (0.9-fold decrease on average) (Figure 6b). Similar expression patterns were observed when we took a close look at the sRNAs derived from different TE contents (Figure S10). Except the circRNA-nonmapped 23nt-sRNAs were slightly increased under drought stress, there was no significant expression difference between mapped and nonmapped sRNAs of other types, no matter with or without drought treatment (Figure S11). Given that the 5′-terminal nucleotides are important for loading of sRNAs onto specific AGO proteins and further their roles (Borges and Martienssen, 2015), we investigated the distributions of the 5′-terminal nucleotides of the circRNA-mapped and -nonmapped sRNAs. As shown in Figure 6(c), 5′-adenine (A) was depleted, while 5′-cytosine (C) was enriched significantly in circRNA-mapped sRNAs as compared with those not mapped to circRNAs (Figure 6c). These results implied that the circRNA-mapped and -nonmapped TE-derived sRNAs, in general, have different AGO preference for loading and may have different roles in regulation of gene expression.

We next asked if the exonic and intronic circRNAs affected the accumulation of sRNAs mapped to their hosting loci. To answer this question, we compared the expression of sRNAs mapped to the whole loci (loci-mapped) and those only mapped to the regions producing circRNAs in these loci (circRNA-mapped) (Figure S12a). Interestingly, we observed that all loci-mapped sRNAs showed significantly higher accumulation than the circRNA-mapped sRNAs, especially the 24nt-sRNAs (Figures 6d and S12b). Taken together, all these data were suggestive of significant correlations between circRNA expression and sRNA accumulation. In most cases, circRNA expression had negative effects on sRNA accumulation.

The expression variations of circRNAs were associated with maize drought tolerance

Association analyses, which are based on accurate phenotypes and extensive coverage by single nucleotide polymorphism (SNP) markers, are powerful tools for dissecting complex agronomic traits, such as flowering time, grain yield, oil biosynthesis and drought tolerance (Huang *et al.*, 2011; Li *et al.*, 2013; Wang *et al.*, 2016). To investigate the possible roles of drought-responsive exonic circRNAs in regulation of maize drought tolerance, we performed candidate gene association analyses on each of these circRNA-mRNA correlated hosting genes in a maize panel consisting of 368 accessions, whose drought-resistant phenotypes and SNP-based genotypes were identified previously (Mao *et al.*, 2015; Wang *et al.*, 2016; Xiang *et al.*, 2017). Using the mixed linear model (MLM) (Zhang *et al.*, 2010), we observed that a large proportion of circRNA-hosting genes showed significant associations with drought tolerance of these maize accessions ($P < 0.05$; Table S14). To test the reliability of these associations, we

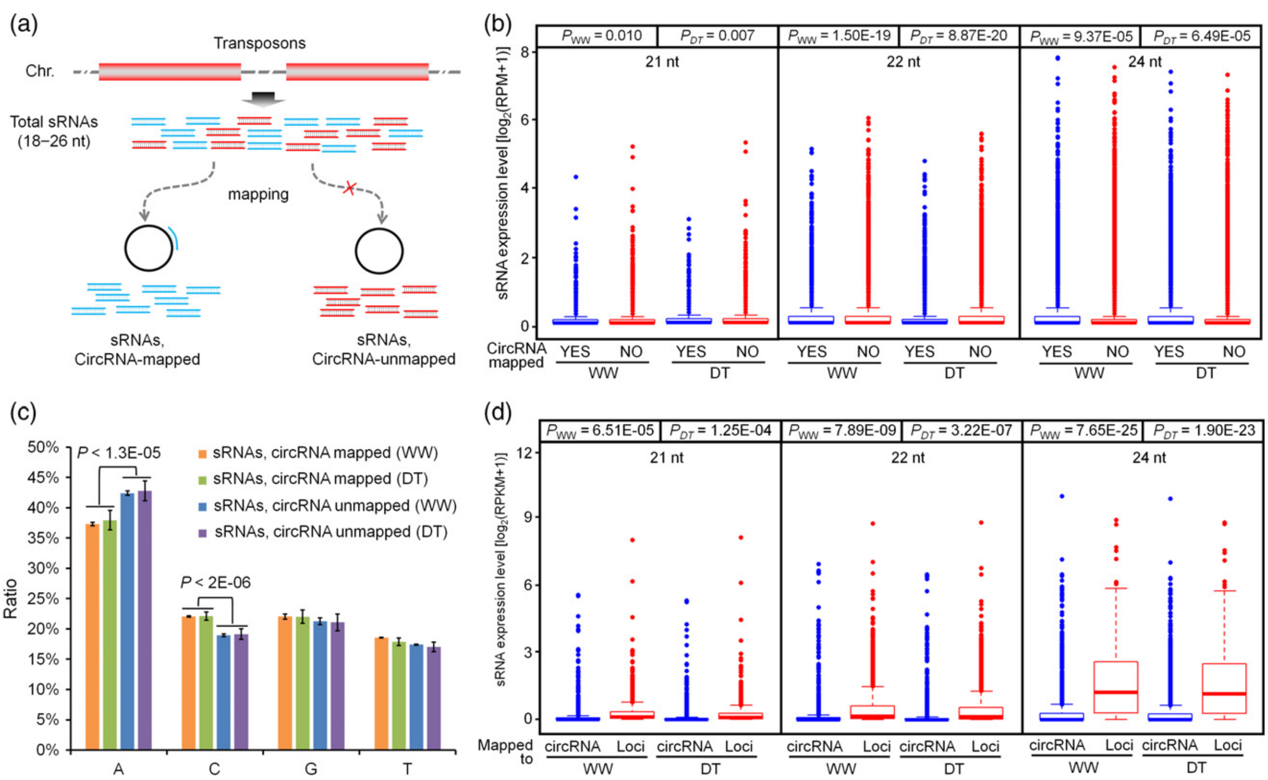

Figure 6. Correlations between circRNA expression and sRNA accumulation.
(a) Schematic diagram indicates the mapping of TE-derived sRNAs to circRNAs.
(b) Expression comparison of 21-nt, 22-nt and 24-nt sRNAs when they were mapped or nonmapped to circRNAs under well watered (WW) or drought (DT) conditions.
(c) The percentages of the 5′-terminal nucleotides in the sRNAs when they were mapped or nonmapped to circRNAs under WW or DT conditions.
(d) Expression comparison of 21-nt, 22-nt and 24-nt sRNAs when they were mapped to the circRNA-hosting loci (loci-mapped) or only to the circRNAs region (circRNA-mapped) under WW or DT conditions.
Statistical significance was calculated using Student's *t*-test.

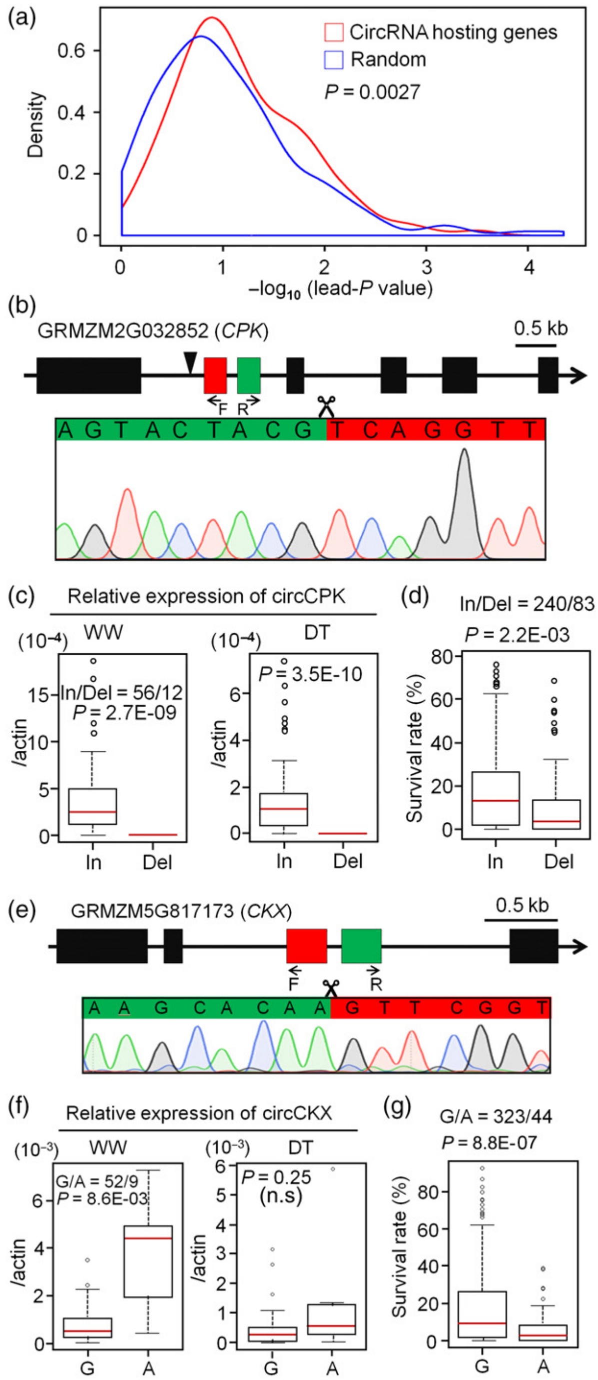

Figure 7. Association analyses of natural variations related to circRNA expression and drought tolerance.
(a) Candidate gene association analyses between genetic variations of cir-RNA-hosting genes and survival rates under drought stress and pairwise random simulations. Density plots show distributions of smallest *P*-values (lead *P*-value) from the candidate genes and the same amounts of randomly selected genes. *t*-tests were performed to compare means of lead *P*-values between candidate and random genes.
(b) Validation of circCPK formation. Red and green colors mark the back-spliced exons. The junction site of circCPK was confirmed by sequencing. The filled boxes indicate the exons and black lines indicate the introns. The black arrowhead indicates the InDel (90 bp) of this gene.
(c) Relative expression levels of circCPK in maize inbred lines (n = 68) with the Insertion (In) or Deletion (Del) variations under WW and DT conditions.
(d) Association analyses between survival rates of a maize association panel (n = 323) after drought stress and Insertion (In) or Deletion (Del) variations in *CPK*.
(e) Validation of circCKX formation. Red and green colors mark the back-spliced exons. The junction site of circCKX was confirmed by sequencing. The filled boxes indicate the exons and black lines indicate the introns.
(f) Relative expression levels of circCKX in maize inbred lines (n = 61) with G or A alleles under WW and DT conditions.
(g) Association analyses between survival rates of the maize panel (n = 367) after drought stress and A or C alleles in *CKX*.
All *P*-values were calculated by *t*-test.

randomly selected same amount genes of the maize genome and did association analyses on them. We collected the smallest *P*-values of the lead SNPs for these random genes and compared their density distributions with those of circRNA-hosting genes. The results showed that the associations identified on the circRNA-hosting genes were more significant than the random tests (Figure 7a), suggesting that the associations detected on these hosting genes are not false positive but real associations.

Next we wanted to know whether variation in expression of circRNAs affects maize drought tolerance. To this end, we investigated the associations between drought tolerance and the expression of several circRNAs in the maize association population. We chose two circRNAs for these experiments: GRMZM2G032852_C1 and GRMZM5G817173_C1. We chose these circRNAs because the orthologs of their parental genes were known to play roles in plant drought tolerance. For example, GRMZM2G032852 encodes a calcium-dependent protein kinase (CPK) family protein, and many plant CPKs have been reported to play positive roles in drought resistance (Ranty *et al.*, 2016). GRMZM5G817173 encodes a cytokinin oxidase/dehydrogenase (CKX), and increased activity of plant CKX greatly enhanced drought tolerance (Pospíšilová *et al.*, 2016). Accordingly, we renamed these circRNAs circCPK and circCKX, respectively.

CircCPK was generated by back-splicing between exons 2 and 3 (Figure 7b). The InDel-932 in the first intron of *CPK* has 90 bp insertion and deletion alleles in the maize panel (Fu *et al.*, 2013). The expression levels of circCPK in plants with the insertion allele were significantly higher than those with the deletion allele under both normal ($P = 2.7 \times 10^{-9}$, n = 68) and drought conditions ($P = 3.5 \times 10^{-10}$) (Figure 7c). After drought stress, the survival rates of the maize plants with the insertion allele were significantly higher than those with the deletion allele ($P = 2.2 \times 10^{-3}$, n = 323) (Figure 7d). CircCKX was formed by back-splicing between exons 3 and 4 (Figure 7e). The SNP chr3.S_191841983 in *CKX* has G and A alleles. Plants with G allele had significantly lower circCKX expression than those with the A allele under normal (P = 0.0086, n = 61) but not drought (P = 0.25) conditions (Figure 7f). The survival rates of the maize plants with the G allele were significantly higher than those with the A allele ($P = 8.8 \times 10^{-7}$, $n = 367$) (Figure 7g). In a parallel expression analysis using RNA-seq data from 368 maize inbreds (Fu *et al.*, 2013), we did not find significant expression difference of *CPK* or *CKX* linear mRNAs between their respective alleles (Figure S13). These results implied that the expression variations of circRNAs might be associated with plant drought tolerance, and that circRNAs could be either effective molecular markers or key players for plant drought tolerance.

Overexpressing circGORK enhanced drought tolerance of Arabidopsis

To validate the circRNA-associated molecular mechanisms in regulation of plant drought tolerance, we overexpressed the circular RNA (circGORK) hosted by the GORK (Guard cell outward-rectifying K$^+$-channel) gene. *GORK* encodes a potassium output channel that plays critical roles in regulation of ABA signaling and water stress responses in Arabidopsis (Hosy *et al.*, 2003). CircGORK was exonic and generated by back-splicing between the second and third exons of *GORK* (Figure 8a). To overexpress circGORK, the genomic DNA fragment containing partial flanking introns (200 bp fragments flanking the second or the third exon), the second exon and the third exon was amplified and inserted after GFP into the pCAMBIA1305-C-3HA binary vector, and the GFP-only binary vector was used as a control (Figure 8b). We observed that circGORK was overexpressed in two independent transgenic lines (Figure 8c). ABA is an important phytohormone that plays pivotal roles in plant stress responses, especially to drought (Pantin *et al.*, 2013; Munemasa *et al.*, 2015). We therefore investigated the roles of circGORK in ABA responses. The results showed that the germination rates of the circGORK OE

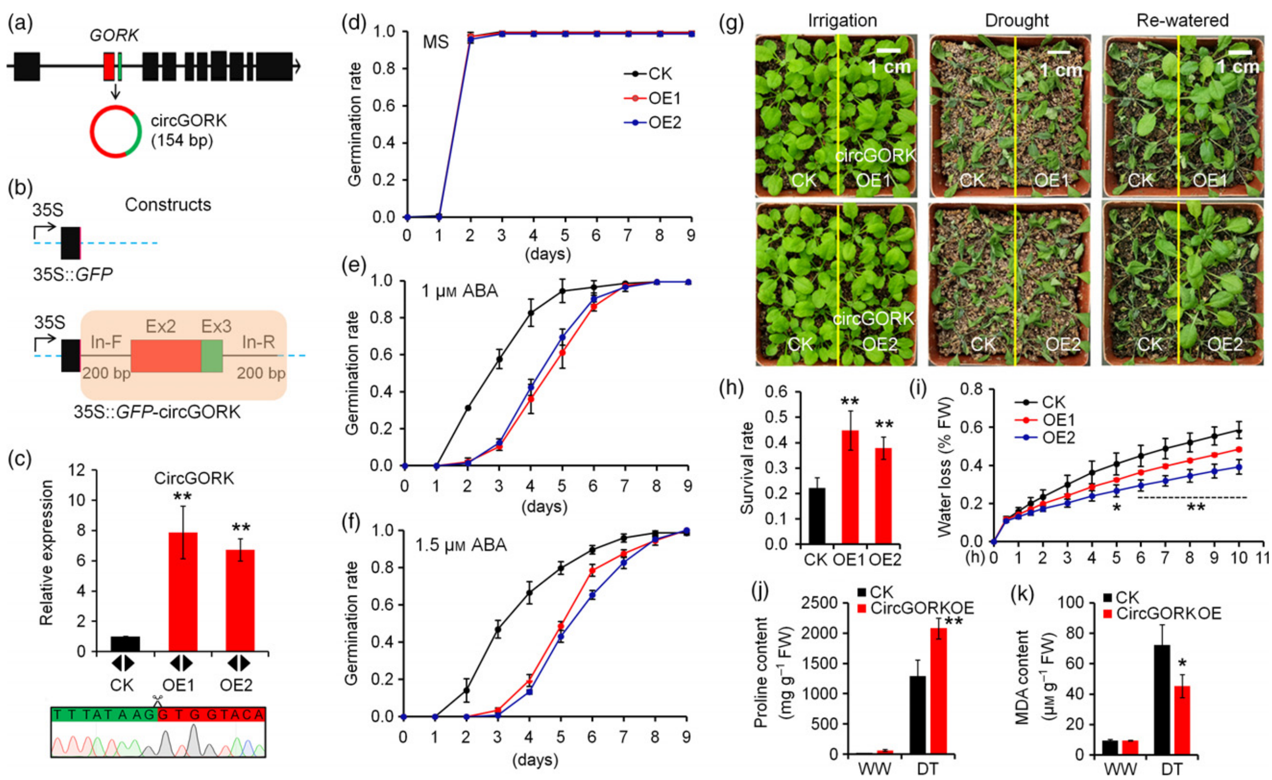

Figure 8. Overexpressing circGORK enhanced Arabidopsis ABA sensitivity and drought tolerance.
(a) CircGORK was exonic and generated by back-splicing between the second and third exons of GORK. The filled boxes indicate the exons and black lines indicate the introns.
(b) Schematic diagram of the constructs overexpressing GFP (35S::GFP, CK) and GFP-circGORK (35S::GFP-circGORK, OE). Exon 2 (Ex2) and exon 3 (Ex3) were inserted into the binary vector with 200 bp of flanking introns. In-F, left intron flanking exon 2; In-R, right intron flanking exon 3. The vertical red line indicates the GFP stop codon.
(c) Expression of circGORK in transgenic CK and OE plants. Data represent the mean ± standard deviation (SD) of three replicates. The back-splicing was confirmed by sequencing of the PCR products with divergent primers (arrowheads indicated). Scissors indicate the circGORK junction site.
(d–f) Germination rates of CK and OE lines on MS plates with or without ABA (1 μM or 1.5 μM). Data represent the mean ± SD of three replicates.
Water loss assay of CK and OE lines.
(g, h) The growth (g) and survival rates (h) of CK and OE lines after drought treatment. Data represent the mean ± SD of three replicates. Bar = 1 cm in (g).
(i) Water loss of detached leaves from CK and transgenic plants at different time points.
(j, k) Proline (j) and MDA (k) contents of CK and OE plants. Data represent the mean ± SD of three replicates.
Significance between difference lines (t-test): $^*P < 0.05$; $^{**}P < 0.01$.

transgenic seeds (OE1, 2) on ABA plates were much lower than those of the control seeds (transformed with empty binary vector), while there was no significant difference in germination rates between control and transgenic seeds on MS plates without ABA (Figure 8d–f; Figure S14). These observations suggested that the transgenic seeds were more sensitive to ABA, and that circGORK played a negative role in ABA responses.

We next grew control and circGORK OE transgenic plants alongside each other in soil. Irrigation was ceased until all leaves were wilted. Then irrigation resumed and the survival rates of the plants were calculated. We observed that the transgenic rather than the control plants survived severe drought stress (Figure 8g,h), suggesting that circGORK OE transgenic plants were more tolerant to drought. We investigated water loss of leaves detached from control and circGORK OE plants and found that circGORK OE leaves lost water more slowly than control leaves (Figure 8i). Further analyses showed that *c.* 1.6 fold higher levels of proline, an osmolyte that protects plants from drought stress damage (Bhaskara *et al.*, 2012), and 37.5% less malondialdehyde (MDA), a metabolic indicator of peroxisomal oxidation of cell membranes (Hernández *et al.*, 2000; Yu *et al.*, 2017), accumulated in circGORK OE plants as compared with controls after drought stress (Figure 8j,k). There were three main isoforms of *GORK* transcripts (ID4, 6 and 7) (Zhang *et al.*, 2017; Figure S15a). We performed expression analysis of ID4, 6 and 7 in circGORK OE plants. In the transgenic plants as compared with the controls, ID6 and ID7 were enhanced under both WW and drought conditions, while ID4 (the most abundant transcript) was slightly enhanced under drought condition (Figure S15b,c). Therefore, overexpression of circGORK was associated with enhanced splicing of ID6 and ID7 under WW conditions and enhanced transcription of the whole gene under drought condition. Taken together, these data indicated a positive role of circGORK in Arabidopsis drought tolerance, and circGORK may have effects on the expression/splicing of its cognate linear transcripts.

Drought promotes the production of ABA, which prevents water loss and protects plant from drought stresses (Pantin *et al.*, 2013; Munemasa *et al.*, 2015). We therefore checked the expression levels of several ABA-responsive marker genes, including *P5CS1*, *RD20*, *RD22*, *RD29A*, *RD29B* and *ABF4*, in circGORK OE plants. The results showed that all of these marker genes were up-regulated in circGORK OE plants after drought stresses or even under WW conditions (Figure S16). Therefore, circGORK could potentially play roles to enhance plant ABA signaling and further promote drought tolerance.

DISCUSSION

Most crops, including primary food crops such as rice and maize, are sensitive to drought. During the past decades, plant scientists had made great progress in characterizing the genetic and molecular mechanisms of plant drought responses. Many regulatory factors and pathways have been identified, such as ABA-dependent and ABA-independent pathways, Ca^{2+} signaling, ROS scavenging molecules including antioxidant enzymes and metabolites, ion channels and non-coding RNAs (Miller *et al.*, 2010; Fujita *et al.*, 2011; Nakashima and Yamaguchi-Shinozaki, 2013; Di *et al.*, 2014; Ferdous *et al.*, 2015; Stephan *et al.*, 2016). Circular RNAs, which form a covalently closed circle, have been recently identified in eukaryotes including plants (Barrett and Salzman, 2016). Whether circRNAs are involved in plant drought responses remains elusive. In this study, we performed high-throughput large-scale expression analyses of circRNAs and mRNAs in both maize and Arabidopsis plants subjected to various levels of drought. We identified a large number of circRNAs in maize and Arabidopsis plants. Consistent with previous studies in animals (Salzman *et al.*, 2013; Guo *et al.*, 2014), we observed that the majority of circRNAs in both maize and Arabidopsis were of low abundance, regardless of whether they were subjected to drought stress. However, expression of numerous circRNAs was upregulated (and a few were downregulated) in both plant species after drought stress (Figures 1 and 5), which suggested that these RNA species could be purposefully produced and have a specific role in drought responses.

Despite their wide roles in regulation of animal development and disease responses (Barrett and Salzman, 2016; Xia *et al.*, 2018), few functions have been ascribed to circRNAs in plants. So far, only a few studies have provided transgenic evidence demonstrating that specific circRNAs could be overproduced or be functional in plant development (Lu *et al.*, 2015; Conn *et al.*, 2017; Tan *et al.*, 2017). In our study, two types of transcriptomes were obtained from maize and Arabidopsis plants subjected to various levels of drought. Thousands of circRNAs and more than 40 000 linear transcripts were obtained in both maize and Arabidopsis. Comparison of the circular and linear transcripts revealed some features of the expression of these two RNA species in response to drought. In maize, the relative numbers of both circRNAs and linear transcripts increased with drought; in Arabidopsis, the overall relative number of circRNAs increased, but the numbers of linear transcripts decreased with drought (Figure 2; Figure S3), suggestive of different response patterns of circRNAs and linear transcripts between maize and Arabidopsis in response to drought. Across various drought treatments, we observed a large proportion of shared linear transcripts in maize and Arabidopsis, but much smaller proportions of shared circRNAs in maize and Arabidopsis (Figure 3; Figure S3). Therefore, the overall expression of circRNAs is more sensitive than that of linear transcripts to drought stress (Figure S6). All these features of circRNAs and linear transcripts in maize and Arabidopsis indicated that these two RNA species are effective genome-wide indicators for plants in drought response.

There are several mechanisms circRNAs use to regulate the linear mRNAs in control of bioprocesses. One is by binding miRNAs, therefore functioning as sponges, to regulate the linear RNAs' expression. A circRNA named ciRS-7, or *CDR1as*, which is produced from the *CDR1* antisense locus, functions as a miRNA sink for miR-7. There are more than 70 miR-7 binding sites on the ciRS-7 molecule, and overexpression of ciRS-7 attenuates the function of miR-7 in cleavage of its target mRNAs (Hansen *et al.*, 2013). In animals, other examples of sponge-like roles of circRNAs in regulation of linear mRNA expression are ciR-SRY, ciR-ITCH, circ-001569 and circRNA-CER (Barrett and Salzman, 2016; Kulcheski *et al.*, 2016). In addition to post-transcriptional regulation, circRNAs can also regulate gene expression at the transcriptional level. For example, a class of exon–intron circRNAs (EIciRNAs), which are located in the nucleus, promote the mRNA expression of their parental genes via binding to the U1 small nuclear ribonucleoprotein component of the spliceosome (Li *et al.*, 2015).

Given the regulation of mRNA expression by circRNAs in animals, we investigated the correlations between circRNAs and their cognate linear transcripts in drought response. About half of the circRNA-hosting genes, with either promoted or inhibited accumulation of circRNAs and mRNAs after drought stress, were enriched in both maize and Arabidopsis (Figure 4). These results suggested correlated expression patterns of circRNAs and mRNAs in response to drought in both species. Because circRNAs have a potential role to regulate their cognate mRNAs in animals, these expression patterns may reflect to some extent regulatory interactions between circRNAs and mRNAs in plants. Indeed, the plant circRNA derived from the *SEP3* gene forms an RNA:DNA hybrid at the parental locus to slow transcriptional elongation, and to recruit the splicing factors and therefore favor the biogenesis of the exon-skipped alternative splicing variants (Conn *et al.*, 2017). Another interesting finding of this study is the negative effects of circRNAs on the accumulation of several types of sRNAs derived from TEs or mapped to the genic regions (Figure 6; Figures S8 and S9). Two scenarios could be expected for how circRNAs regulate sRNA accumulation: (i) circRNAs could play a sponge role; (ii) circRNAs could inhibit the process of sRNA biogenesis. In the future, it will be important to further study the mechanisms by which plant circRNAs regulate their cognate linear RNAs or correlated sRNAs in response to drought, and our study has provided a list of candidate targets for these studies.

Due to the lack of 5′ and 3′ ends, the circular RNAs are stable as they are not degraded as efficiently as linear RNAs (Jeck *et al.*, 2013). Therefore, stable circRNAs have the potential to be used as molecular markers. Indeed, some circRNAs are only expressed in certain disease conditions or specific cell types in mammals. It was reported that hsa_circ_002059, a circRNA significantly downregulated in gastric cancer tissues, was used as a stable biomarker for diagnosis of gastric carcinoma (Li *et al.*, 2015). Our results uncovered a group of plant circRNAs that were differentially expressed under certain drought stress conditions in both maize and Arabidopsis (Figure 2). Additionally, by using a maize association population, we found that the variation in expression of certain circRNAs, exemplified by circCPK and circR_CKX, were significantly associated with maize drought tolerance. Therefore, circRNAs have the potential to be used as molecular markers in breeding drought-tolerant crop cultivars.

EXPERIMENTAL PROCEDURES

Plant materials and drought treatments

Maize (*Zea mays* L.) inbred line B73 and Arabidopsis ecotype Col-0 were used in this study. B73 seeds were directly sown in soil at 28°C in black pots (10 cm × 10 cm × 9 cm) with 16 h light (*c.* 140 μmol m^{-2} s^{-1})/8 h dark in a glasshouse. Col-0 seeds were stratified in a cold room for 2 days at 4°C in the dark, germinated on culture medium for 7 days and then transferred to soil to grow at 22°C with 16 h light (*c.* 85 μmol m^{-2} s^{-1})/8 h dark in a growth chamber.

For drought experiments, gradient drought treatments were administered 16 days after planting maize and 21 days after planting Arabidopsis according to a previous report (Su *et al.*, 2013). Briefly, the soil moisture of each treatment was monitored by measuring the relative soil water content, followed by rationing of water to maintain a designated soil moisture. The pots were weighed and watered twice daily. After the most serious stress reached the preset levels, the plants were maintained for an extra 1 day, then the whole plants were harvested for fresh weight measurement and leaves were harvested for RNA extraction. The third leaves of B73 at V4 stage from at least three plants and leaves of four Col-0 plants were harvested and pooled, respectively, for RNA extraction. Two biological replicates were collected for each sample. Leaf samples were frozen in liquid nitrogen right after sampling and then stored at −80°C until RNA extraction.

For the experiments on maize inbred lines, two treatments (WW, 65–60% soil moisture; drought-stressed (DT), 25–20% soil moisture) were applied. Eight plants of each inbred line were grown in two pots (10 cm × 10 cm × 9 cm), with four plants in each pot. All pots were randomly placed in the growth room at 28°C with 16 h light/8 h dark. Soil moisture was controlled by measuring the relative water content of soil and rationing water (pots were weighed and watered once daily). Drought treatments started 16 days after planting, then after 4 days of drought treatment leaves were sampled. The uppermost expanded leaves, most of which were third leaves at the V4 stage, were harvested for RNA extraction. For most samples, two replicates were obtained, and each replicate contained at least three leaves from one pot.

Total RNA and genomic DNA extraction

Total RNAs were isolated from the pooled samples using TRIpure reagent (Aidlab, Beijing, CN) according to the manufacturer's protocol. Total RNA samples were treated with DNase I (Thermo

Scientific) and purified by RNA Clean & Concentrator-25 spin columns (ZYMO Research). The quality of RNA was evaluated using 2% TAE agarose gel electrophoresis. Genomic DNA from maize was extracted with conventional cetyltrimethylammonium bromide (CTAB). For RNase *R* treatment, purified DNase I-treated total RNA was incubated for 15 min at 37°C with 3 units RNase *R* per μg of total RNA (Epicentre, Shanghai, CN). RNAs were subsequently purified with the sodium acetate-ethanol solution. RNA samples treated with or without RNase *R* were reverse transcribed with random primers using SuperScript II reverse transcriptase (Invitrogen, Shanghai, CN).

CircRNA-sequencing analyses

RNA samples from maize and Arabidopsis treated with various levels of drought were digested with RNase *R* before library preparation and then subjected to deep sequencing using an Illumina HiSeq 3000 on a 150 bp paired-end run (Ribobio, Guangzhou, CN). Clean FASTQ reads were first mapped to the maize B73 reference genome from EnsemblePlants (ftp://ftp.ensemblgenomes.org/pub/plants/release-31/fasta/zea_mays/dna/) or Arabidopsis genome (TAIR10.dna_sm.toplevel.fa) obtained from the public database (ftp://ftp.ensemblgenomes.org/pub/plants/release-32/fasta/arabidopsis_thaliana/dna/) by BWA-MEM with a CIRI2 suggested parameter: T, 19 (Li and Durbin, 2009; Gao *et al.*, 2015, 2018). SAM alignment files generated by BWA-MEM with GTF-formatted gene annotations (Zea_mays.AGPv3.31.gtf, Arabidopsis_thaliana.TAIR10.32.gtf) were then called by the CIRI2 algorithm with default parameters except thread number to detect and annotate circRNAs. All CIRI2 circRNA outputs were supported by more than two junction reads (reads spanning back-spliced sites).

CircRNAs with at least two junction reads that occurred in both replicates were selected as high-confidence predictions. They were then compared with circRNAs in PlantcircBase (http://ibi.zju.edu.cn/plantcircbase/) (Chu *et al.*, 2017, 2018) to identify different circRNAs discovered in this study. To show the whole scenario of circRNA variation under various degree of drought stresses, we counted the number of distinct circRNAs in each treatment and normalized the number by total clean reads (units in million), and grouped circRNAs based on genomic origin type.

The number of junction reads was normalized by total number of cleaned sequencing reads (units in billions) to estimate the relative expression of a circRNA, denoted as RPB (junction reads per billion clean reads). RPB datasets were qualified by visualizing results of PCA. CircRNAs were classified by relative expression levels: low (<100 RPB), medium (100–500 RPB), high (500–1000 RPB) or extremely high (≥1000 RPB). To characterize circRNAs with higher expression levels, the exon length and flanking-intron length of circRNA junction sites were calculated based on gene annotations (*Zea_mays*.AGPv3.31.gff3, *Arabidopsis_thaliana*.TAIR10.31.gff3).

Reads per billion datasets were used to call differentially expressed circRNAs (DE circRNAs) from different treatments using the Bioconductor package edgeR. The edgeR programs were performed in OmicShare tools, a free online platform for data analysis, with default parameters (www.omicshare.com/tools). The cutoff we used to pick significant DE circRNAs was FDR < 0.05. The expression of DE circRNAs was depicted by a heatmap generated by pheatmap in R with parameters: scale, 'row', cluster_cols, F.

RNA sequencing

The same RNA samples used in circRNA-seq were used for library preparation and RNA-seq. 150 bp paired-end raw reads were generated using an Illumina HiSeq3000 (ANOROAD, Beijing, CN). Clean reads (*c.* 97% of the total raw reads) were first mapped to the transcriptome of maize B73 (Zea_mays.AGPv3.31.cdna.all.fa) or Arabidopsis Col (Arabidopsis_thaliana.TAIR10.cdna.all.fa) obtained from the Ensemble genome database (ftp://ftp.ensemblgenomes.org/pub/plants/release-31/fasta/zea_mays/cdna/, ftp://ftp.ensemblgenomes.org/pub/plants/release-32/fasta/arabidopsis_thaliana/cdna/) by rsem-prepare-reference –bowtie with default parameters (Dewey and Li, 2011). The average percentage of reads with at least one reported alignment was 82.95% in maize and 92.87% in Arabidopsis. Transcript quantification values were then calculated by rsem-calculate-expression with the following parameters: num-threads, 8; calc-pme; calc-ci. The TPM (transcripts per million) values were selected to represent relative expression of a transcript.

Differential expression analyses were performed on TPM datasets by edgeR in the OmicShare tools to identify differentially expressed linear RNAs (DE mRNAs). Transcripts with FDR < 0.05 and |Fold Change|≥2 were considered to be significant DE mRNAs. Fold change of DE circRNAs and DE mRNAs were showed by violin plot to compare change intensity of the two kinds of RNA species in response to drought.

Small RNA data processing and expression calculation

The sRNA data were downloaded from EMBL ArrayExpress (E-GEOD-70487, https://www.ebi.ac.uk/arrayexpress/experiments/E-GEOD-70487) Poor-quality (Q20 ≤ 50% or N ≥ 10%) and low complexity (A&T content >80% or poly A/T content >30%) reads were removed using custom Perl script. Reads shorter than 18 nt or longer than 26 nt were excluded from further analysis. Reads mapped to maize tRNA, rRNA, snRNA, snoRNA obtained from GtRNAdb (http://lowelab.ucsc.edu/GtRNAdb/Zmays5/), Rfam (ftp://ftp.ebi.ac.uk/pub/databases/Rfam/12.0/), plant snoRNA database (http://bioinf.scri.sari.ac.uk/cgi-bin/plant_snorna/home/), silva (http://www.arb-silva.de/) were also removed. The remained reads were mapped to the maize genome with no mismatch allowed. All reads mapped to genomic sequence were further mapped to the TE sequences collected from maize TE database (http://maizetedb.org/~maize/), circRNA-hosting loci and cirRNA regions identified in this study. The expression levels of sRNAs were measured by RPM (read per million mapped) or RPKM (read per kilobase per million mapped) and calculated by custom Perl script.

Expression clustering

DECs (genes that produce differentially expressed circRNAs) and DELs (genes that produce differentially expressed linear RNAs) were compared with designate DEGs that differentially expressed both DE circRNAs and DE mRNAs. Million-fold random simulations were conducted to confirm the significance of genomic loci coincidences. Randomly selected numbers as DECs and DELs at all protein coding gene (maize, 39 469; Arabidopsis, 27 628) backgrounds were compared and the probability that the number of intersections was not smaller than the number of DEGs was calculated.

DEG expression pattern analyses were performed by Short Time-series Expression Miner software (STEM) on the OmicShare tools platform with parameters set as follows: maximum unit change in model profiles between time points was 1, maximum output profiles number was 20 and minimum ratio of fold change of DEGs was no less than 2.0. DE circular and linear transcripts were assigned to 20 expression patterns. Then, DEGs with single up (C+ or L+, C means circRNAs, L means linear RNAs) or single down (C- or L-) trends during increasingly severe drought treatments were gathered to four groups named optimal expression patterns (C+L+, C-L-, C+L-, C-L+). Additionally, the DEG preference for the optimal pattern was revealed by random simulations. For each optimal pattern, two sets of randomly selected DEGs were constructed based on total number of DEGs, one of which expressed circRNAs in the optimal trend while the other expressed mRNAs in the optimal trend. The number of the intersection of the two sets was counted and compared with the number of genes in the optimal pattern. Each simulation was performed one million times and the probability that DEGs fell into the optimal pattern was calculated.

Validation of circRNAs

PCR primers (divergent and convergent) were designed for circRNA validation. cDNA (RNase R+), genomic DNA (gDNA) and total RNA were used as templates for the PCR validation of circRNAs. The reagent 2× Taq Master Mix (Vazyme, Nanjing, CN) was used for cDNA and gDNA amplification with touchdown PCR to detect the circRNA templates. Touchdown PCR was carried out using a Bio-RadT100 Thermocycler with our customized program (95°C × 3 min for one cycle; 95°C × 30 sec, 60°C × 30 sec, 72°C × 30 sec, for nine cycles by decreasing 0.5°C per cycle; 95°C × 30 sec, 55°C × 30 sec, 72°C × 30 sec, for 30 cycles, 72°C × 5 min for one cycle, 12°C × 1 sec for one cycle). Then, Sanger sequencing was performed on all PCR products. Relative quantitative PCR (QPCR) reactions were used to validate the expression of circular and linear transcripts.

KEGG analyses

To profile drought-responsive mechanisms, enrichment analyses of biological pathways defined by KEGG were conducted. The DE circRNAs and DE mRNAs from maize and Arabidopsis were first converted and filtered to Entrez Gene IDs by customized scripts according to gene information (*Arabidopsis_thaliana*.gene_info, *Zea_mays*.gene_info) from GenBank. Then, the Entrez IDs were called by the Gene-list Enrichment tool in KOBAS3.0 to do KEGG pathway enrichment with default parameters (Xie *et al.*, 2011). The cut-off for significant pathways was set as $P < 0.01$.

Constructs and transgenic studies

To overexpress circGORK, flanking introns (200 bp) and exons were cloned from genomic DNA and cDNA with Phanta Super-Fidelity (Vazyme) respectively; then, these PCR fragments were joined by a fusion-PCR method. This fused fragment was then cloned into the *Bam*HI/*Xba*I sites of pRTL2–EGFP. A fragment containing the circGORK sequence, TEV Leader and EGFP from pRTL2–egfp-circGORK was then amplified and inserted into pCAMBIA1305-C-3HA by Exnase II-mediated recombination (Vazyme). Transgenic *Arabidopsis thaliana* lines were generated by the *Agrobacterium*-mediated floral dip transformation method (Clough and Bent, 1998), and homozygous transgenic T3 lines carrying a single T-DNA insertion were used for further analyses.

Measurements of proline and malondialdehyde (MDA) contents

These experiments were performed essentially according to a previous report (He *et al.*, 2018). Briefly, well watered and drought-treated seedlings were used for proline and MDA measurements. For proline measurement, 0.1 g leaves were ground with a mortar and pestle in liquid nitrogen then transferred to a 10 ml tube. Proline was then extracted with 4 ml 3% sulphosalicylic acid by placing the reaction mixture in a boiling water bath for 30 min and cooling to room temperature. One ml 2.5% acid-ninhydrin and 1 ml acetic acid were added to 1 ml supernatant and boiled for another 30 min. Four ml methylbenzene were added into the mixture for extracting the proline. Proline extracts were centrifuged at 13 500 ***g*** for 15 min to remove impurities, then the absorbance of the supernatant at 520 nm was determined with a NanoDrop200 bioanalyser (Thermo Scientific). The proline concentrations of the tested samples were determined by extrapolation against a standard curve of L-proline.

For MDA measurement, 1.7 ml 5% trichloroacetic acid (TCA) was added to 80 mg ground leaves in a 2 ml tube, and the mixture was centrifuged at 850 ***g*** for 10 min at 4°C. One ml supernatant was added to 1 ml 0.67% thiobarbituric acid (TBA) and the mixture was incubated in a water bath for 30 min (*c.* 95°C). The absorbances of the supernatant at 450, 532 and 600 nm were measured. The MDA concentration was determined as C μmol^{-1} $L^{-1} = 6.45*(A_{532} - A_{600}) - 0.56*A_{450}$.

Germination assays

Seeds were surface sterilized with 10% sodium hypochlorite solution and incubated for 3 days at 4°C in the dark. Then, the seeds were sown on Petri dishes containing 0.44% (w/v) Murashige and Skoog (MS) Basal Medium (Sigma), 0.05% (w/v) 2-(*N*-morpholino) ethanesulfonic acid (MES; Sigma), 1% (w/v) sucrose (SCR), 0.3% phytagel (Sigma) with or without ABA (1 μM and 1.5 μM) and placed in growth chambers under 16 h light/8 h dark conditions at 22°C. The germination rates were investigated daily under the microscope.

Drought treatment and water loss measurements of transgenic plants

For drought treatment, seedlings grown on MS medium for 7 days were transplanted to soil (CK and circGORK-overexpression lines were planted side by side) and well watered for 10 days, then stopped watering for about 2 weeks, and investigated the survival rates after re-watering. For water loss measurement, rosette leaves of 3-week-old CK and circGORK-overexpression plants were detached and weighed at various time intervals. Water loss was expressed as the percentage of initial fresh weight.

Association analyses

Association analyses for maize circRNA-hosting genes were performed using a maize association mapping population consisting of 368 inbred lines, whose drought-tolerant phenotypic data were published previously (Mao *et al.*, 2015; Wang *et al.*, 2016; Xiang *et al.*, 2017). The SNPs used in these analyses have minor allele frequencies (MAF) ≥ 0.05. An MLM was used to detect significantly associated SNPs by using the TASSEL software (Bradbury *et al.*, 2007). The distributions of lowest *P*-values (lead *P*-value) from circRNA-hosting genes and the same amount of randomly selected genes were presented by density plot and the mean values were compared by Student's *t*-test.

ACCESSION NUMBER

All the circRNA-seq and RNA-seq data can be publicly reached in NCBI GEO with accession number: GSE124340.

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CONFLICT OF INTERESTS

The authors declare no conflicts of interest.

AUTHORS' CONTRIBUTIONS

YF and PZ were involved in plant growth and drought treatment experiments. PZ conducted the molecular experiments. YF, XS and LL analyzed the data. MD designed the experiments. MD, LL and WT wrote the manuscript. All authors read and approved the final manuscript.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Pipeline of comparative transcriptomic analysis.

Figure S2. Principle component analyses (PCA) of transcriptomic datasets.

Figure S3. mRNA dynamics across drought treatments.

Figure S4. Genes with higher expression levels were more likely to produce circRNAs.

Figure S5. Classification of 73 validated circRNAs.

Figure S6. Validation of the expression patterns of maize circRNAs and their cognate mRNAs in response to drought.

Figure S7. Fold changes of plant circRNA and mRNA transcripts in response to drought.

Figure S8. Expression patterns of ortholog DE circular RNAs.

Figure S9. Examples of co-detected drought-responsive pathways in Arabidopsis and maize.

Figure S10. Expression of three major sRNAs from different TE contents.

Figure S11. Expression comparison between circRNA-mapped and circRNA-nonnonmapmapped sRNAs.

Figure S12. Expression comparison between loci-mapped and circRNA-mapped sRNAs.

Figure S13. The relationship between the nature variations and the circRNA-hosting gene expression.

Figure S14. Germination phenotypes of CK and circGORK overexpressing lines.

Figure S15. Expression levels of three GORK variants.

Figure S16. Overexpression of circGORK enhanced the expression of ABA-responsive marker genes.

Table S1. Summary of circRNA-seq data for maize plants grown in various drought environments.

Table S2. Summary of circRNA-seq data for Arabidopsis plants grown in various drought environments.

Table S3. Summary of RNA-seq data for maize plants grown in various drought environments.

Table S4. Summary of RNA-seq data for Arabidopsis plants grown in various drought environments.

Table S5. Expression of 2174 high-confident maize circRNAs expressed after subjections to various levels of drought stress.

Table S6. Expression of 1354 high-confident Arabidopsis circRNAs.

Table S7. Expression of linear transcripts in maize treated with various levels of drought stress.

Table S8. Expression of linear transcripts in Arabidopsis treated with various levels of drought stress.

Table S9. Summary of validated circRNAs and primers used in this study.

Table S10. Collection of 1843 maize circRNAs differentially expressed after subjections to various levels of drought stress.

Table S11. Collections of 8681 maize linear transcripts differentially expressed after subjection to various levels of drought stress.

Table S12. Collections of 1283 Arabidopsis circRNAs differentially expressed after subjection to various levels of drought stress.

Table S13. Collection of 9322 Arabidopsis linear transcripts differentially expressed after subjection to various levels of drought stress.

Table S14. Association of genetic polymorphisms in circRNA-hosting genes with drought tolerance at seedling stage in 368 maize inbred lines.

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